

 INAF ISTITUTO NAZIONALE DI ASTRONOMIA OSSERVATORIO ASTRONOMICO DI ARCETRI	FATE MONTHLY REPORT Doc.No : FATE-001 Version : 1.0 Date : 03-09-2025	 CONSORZIO LaMMA
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FORECAST SYSTEM FOR ATMOSPHERE AND TURBULENCE at the VERY LARGE TELESCOPE

Performance assessment
Annex
August 2025



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1 NIGHT1: Contingency tables

Tables in section 1 are included in the range: 1-22.

wind speed ms^{-1}		Observations		
		ws < 4.2	4.2 < ws < 7.6	ws > 7.6
Model data	ws < 4.2	5962	2279	336
	4.2 < ws < 7.6	3641	4855	1585
	ws > 7.6	428	2123	8228
Number of points 29437 ; PC=64.7%; EBD=2.6%; POD1=59.4%; POD2=52.4%; POD3=81.1%				

Table 1: night1 - Contingency table for wind speed (ms^{-1}) in standard configuration

wind speed		Observations		
ms^{-1}		ws < 4.2	4.2 < ws < 7.6	ws > 7.6
Model data	ws < 4.2	9355	647	0
	4.2 < ws < 7.6	676	8133	509
	ws > 7.6	0	477	9640
Number of points 29437 ; PC=92.2%; EBD=0.0%; POD1=93.3%; POD2=87.9%; POD3=95.0%				

Table 2: night1 - Contingency table for wind speed (ms^{-1}) processed with AR (1H)

wind speed	Observations			
ms^{-1}	ws < 12.0	12.0 < ws < 18.0	ws > 18.0	
Model data	ws < 12.0	25766	264	0
	12.0 < ws < 18.0	211	2634	63
	ws > 18.0	0	76	423
Number of points 29437 ; PC=97.9%; EBD=0.0%; POD1=99.2%; POD2=88.6%; POD3=87.0%				

Table 3: night1 - Contingency table for wind speed (ms^{-1}) processed with AR (1H) and thresholds (12.0,18.0)

FATE - Monthly Report

wind direction degree		North	East	South	Observations	West
Model data	North	13250	801	3		176
	East	1189	2054	458		10
	South	129	534	3317		164
	West	472	117	21		227
Number of points 22922 ; PC=82.2%; EBD=1.1%; POD1=88.1%; POD2=58.6%; POD3=87.3%; POD4=39.3%						

Table 4: night1 - Contingency table for wind direction (degree) in standard configuration

wind direction degree		North	East	South	Observations	West
Model data	North	14765	204	0		56
	East	249	3191	98		0
	South	0	111	3689		16
	West	26	0	12		505
Number of points 22922 ; PC=96.6%; EBD=0.0%; POD1=98.2%; POD2=91.0%; POD3=97.1%; POD4=87.5%						

Table 5: night1 - Contingency table for wind direction (degree) processed with AR (1H)

relative humidity percent		Observations		
		rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	6123	2070	11
	7.1 < rh < 15.3	3965	5886	1133
	rh > 15.3	479	2831	6610
Number of points 29108 ; PC=64.0%; EBD=1.7%; POD1=57.9%; POD2=54.6%; POD3=85.2%				

Table 6: night1 - Contingency table for relative humidity (percent) in standard configuration

relative humidity percent		Observations		
		rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	9968	466	3
	7.1 < rh < 15.3	598	9987	207
	rh > 15.3	1	334	7544
Number of points 29108 ; PC=94.5%; EBD=0.0%; POD1=94.3%; POD2=92.6%; POD3=97.3%				

Table 7: night1 - Contingency table for relative humidity (percent) processed with AR (1H)

relative humidity percent		Observations		
		rh < 50.0	50.0 < rh < 70.0	rh > 70.0
Model data	rh < 50.0	28453	49	0
	50.0 < rh < 70.0	54	483	9
	rh > 70.0	0	7	53
Number of points 29108 ; PC=99.6%; EBD=0.0%; POD1=99.8%; POD2=89.6%; POD3=85.5%				

Table 8: night1 - Contingency table for relative humidity (percent) processed with AR (1H) and thresholds (50,70) percent

precipitable water vapor mm		Observations		
		pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	7816	1561	10
	1.9 < pwv < 3.5	1189	3837	726
	pwv > 3.5	62	833	5971
Number of points 22005 ; PC=80.1%; EBD=0.3%; POD1=86.2%; POD2=61.6%; POD3=89.0%				

Table 9: night1 - Contingency table for precipitable water vapor (mm) in standard configuration

precipitable water vapor mm		Observations		
		pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	8922	206	0
	1.9 < pwv < 3.5	145	5953	87
	pwv > 3.5	0	72	6620
Number of points 22005 ; PC=97.7%; EBD=0.0%; POD1=98.4%; POD2=95.5%; POD3=98.7%				

Table 10: night1 - Contingency table for precipitable water vapor (mm) processed with AR (1H)

FATE - Monthly Report

total seeing arcsec	Observations		
	see < 0.6	0.6 < see < 0.8	see > 0.8
see < 0.6	3252	2771	1029
Model data 0.6 < see < 0.8	2361	2865	2328
see > 0.8	379	954	2568

Number of points 18507 ; PC=46.9%; EBD=7.6%; POD1=54.3%; POD2=43.5%; POD3=43.3%

Table 11: night1 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.0

total seeing arcsec	Observations		
	see < 0.6	0.6 < see < 0.8	see > 0.8
see < 0.6	5975	988	34
Model data 0.6 < see < 0.8	803	4998	591
see > 0.8	6	680	5444

Number of points 18507 ; PC=84.1%; EBD=0.2%; POD1=88.1%; POD2=75.0%; POD3=89.7%

Table 12: night1 - Contingency table for total seeing (arcsec) processed with AR (1H) and accuracy 0.0

total seeing arcsec	Observations		
	see < 0.6	0.6 < see < 0.8	see > 0.8
see < 0.6	5707	0	738
Model data 0.6 < see < 0.8	0	6451	738
see > 0.8	285	139	5187

Number of points 18507 ; PC=93.7%; EBD=5.5%; POD1=95.2%; POD2=97.9%; POD3=77.8%

Table 13: night1 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.24

total seeing arcsec	Observations		
	see < 0.6	0.6 < see < 0.8	see > 0.8
see < 0.6	6778	25	30
Model data 0.6 < see < 0.8	0	6620	29
see > 0.8	6	21	6039

Number of points 18507 ; PC=99.6%; EBD=0.2%; POD1=99.9%; POD2=99.3%; POD3=99.0%

Table 14: night1 - Contingency table for total seeing (arcsec) processed with AR (1H) and accuracy 0.24

FATE - Monthly Report

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	2648	1756	130
	$3.4 < \tau < 5.8$	967	3270	1779
	$\tau > 5.8$	200	1717	4561
Number of points 17028 ; PC=61.5%; EBD=1.9%; POD1=69.4%; POD2=48.5%; POD3=70.5%				

Table 15: night1 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 0.0

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	3516	328	4
	$3.4 < \tau < 5.8$	299	6021	473
	$\tau > 5.8$	0	394	5993
Number of points 17028 ; PC=91.2%; EBD=0.0%; POD1=92.2%; POD2=89.3%; POD3=92.6%				

Table 16: night1 - Contingency table for wavefront coherence time (ms) processed with AR (1H) and accuracy 0.0

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	3340	223	697
	$3.4 < \tau < 5.8$	275	5568	697
	$\tau > 5.8$	200	952	5773
Number of points 17028 ; PC=86.2%; EBD=5.3%; POD1=87.5%; POD2=82.6%; POD3=80.5%				

Table 17: night1 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 1.22

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	3811	2	36
	$3.4 < \tau < 5.8$	4	6694	36
	$\tau > 5.8$	0	47	6434
Number of points 17028 ; PC=99.5%; EBD=0.2%; POD1=99.9%; POD2=99.3%; POD3=98.9%				

Table 18: night1 - Contingency table for wavefront coherence time (ms) processed with AR (1H) and accuracy 1.22

FATE - Monthly Report

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	2582	2786
	0.6 < glf < 0.7	1875	2411
	glf > 0.7	376	946
Number of points 17224 ; PC=37.4%; EBD=14.0%; POD1=53.4%; POD2=39.2%; POD3=23.3%			

Table 19: night1 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.0

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	4084	889
	0.6 < glf < 0.7	715	4377
	glf > 0.7	34	877
Number of points 17224 ; PC=79.3%; EBD=0.4%; POD1=84.5%; POD2=71.3%; POD3=83.1%			

Table 20: night1 - Contingency table for ground layer fraction (dimensionless) processed with AR (1H) and accuracy 0.0

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	4398	669
	0.6 < glf < 0.7	59	5442
	glf > 0.7	376	32
Number of points 17224 ; PC=80.4%; EBD=15.2%; POD1=91.0%; POD2=88.6%; POD3=49.4%			

Table 21: night1 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.14

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	4785	51
	0.6 < glf < 0.7	14	6073
	glf > 0.7	34	19
Number of points 17224 ; PC=99.0%; EBD=0.5%; POD1=99.0%; POD2=98.9%; POD3=98.6%			

Table 22: night1 - Contingency table for ground layer fraction (dimensionless) processed with AR (1H) and accuracy 0.14

2 DAY1: Contingency tables

Tables in section 2 are included in the range: 23-32.

wind speed ms^{-1}		Observations		
		ws < 4.6	4.6 < ws < 7.4	ws > 7.4
Model data	ws < 4.6	5311	1614	348
	4.6 < ws < 7.4	4159	5644	1833
	ws > 7.4	388	2302	7863
Number of points 29462 ; PC=63.9%; EBD=2.5%; POD1=53.9%; POD2=59.0%; POD3=78.3%				

Table 23: day1 - Contingency table for wind speed (ms^{-1}) in standard configuration

wind speed		Observations		
ms^{-1}		ws < 4.6	4.6 < ws < 7.4	ws > 7.4
Model data	ws < 4.6	8815	737	8
	4.6 < ws < 7.4	1038	8327	462
	ws > 7.4	5	496	9574
Number of points 29462 ; PC=90.7%; EBD=0.0%; POD1=89.4%; POD2=87.1%; POD3=95.3%				

Table 24: day1 - Contingency table for wind speed (ms^{-1}) processed with AR (1H)

wind speed	Observations			
ms^{-1}	ws < 12.0	12.0 < ws < 18.0	ws > 18.0	
	ws < 12.0	25194	308	0
Model data	12.0 < ws < 18.0	224	2944	88
	ws > 18.0	0	95	609
Number of points 29462 ; PC=97.6%; EBD=0.0%; POD1=99.1%; POD2=88.0%; POD3=87.4%				

Table 25: day1 - Contingency table for wind speed (ms^{-1}) processed with AR (1H) and thresholds (12.0,18.0)

FATE - Monthly Report

wind direction degree		North	East	South	Observations	West
Model data	North	10854	886	10		258
	East	213	740	111		1
	South	49	63	1748		628
	West	3426	42	281		6208
Number of points 25518 ; PC=76.6%; EBD=0.4%; POD1=74.6%; POD2=42.7%; POD3=81.3%; POD4=87.5%						

Table 26: day1 - Contingency table for wind direction (degree) in standard configuration

wind direction degree		North	East	South	Observations	West
Model data	North	13999	122	2		313
	East	130	1581	10		0
	South	0	26	2030		92
	West	413	2	108		6690
Number of points 25518 ; PC=95.2%; EBD=0.0%; POD1=96.3%; POD2=91.3%; POD3=94.4%; POD4=94.3%						

Table 27: day1 - Contingency table for wind direction (degree) processed with AR (1H)

relative humidity percent		Observations		
		rh < 8.9	8.9 < rh < 17.9	rh > 17.9
Model data	rh < 8.9	7447	1825	39
	8.9 < rh < 17.9	3189	6787	1694
	rh > 17.9	218	2176	6027
Number of points 29402 ; PC=68.9%; EBD=0.9%; POD1=68.6%; POD2=62.9%; POD3=77.7%				

Table 28: day1 - Contingency table for relative humidity (percent) in standard configuration

relative humidity percent		Observations		
		rh < 8.9	8.9 < rh < 17.9	rh > 17.9
Model data	rh < 8.9	10378	460	1
	8.9 < rh < 17.9	476	10033	239
	rh > 17.9	0	295	7520
Number of points 29402 ; PC=95.0%; EBD=0.0%; POD1=95.6%; POD2=93.0%; POD3=96.9%				

Table 29: day1 - Contingency table for relative humidity (percent) processed with AR (1H)

relative humidity percent		Observations		
		rh < 50.0	50.0 < rh < 70.0	rh > 70.0
Model data	rh < 50.0	28592	36	0
	50.0 < rh < 70.0	35	648	10
	rh > 70.0	0	2	79
Number of points 29402 ; PC=99.7%; EBD=0.0%; POD1=99.9%; POD2=94.5%; POD3=88.8%				

Table 30: day1 - Contingency table for relative humidity (percent) processed with AR (1H) and thresholds (50,70) percent

FATE - Monthly Report

precipitable water vapor mm	Observations			
		pwv < 2.1	2.1 < pwv < 3.7	pwv > 3.7
Model data	pwv < 2.1	8062	1287	0
	2.1 < pwv < 3.7	1534	4051	836
	pwv > 3.7	87	966	5994
Number of points 22817 ; PC=79.4%; EBD=0.4%; POD1=83.3%; POD2=64.3%; POD3=87.8%				

Table 31: day1 - Contingency table for precipitable water vapor (mm) in standard configuration

precipitable water vapor mm	Observations			
		pwv < 2.1	2.1 < pwv < 3.7	pwv > 3.7
Model data	pwv < 2.1	9474	111	0
	2.1 < pwv < 3.7	209	6081	111
	pwv > 3.7	0	112	6719
Number of points 22817 ; PC=97.6%; EBD=0.0%; POD1=97.8%; POD2=96.5%; POD3=98.4%				

Table 32: day1 - Contingency table for precipitable water vapor (mm) processed with AR (1H)

3 NIGHT2: Contingency tables

Tables in section 3 are included in the range: 33-42.

wind speed ms^{-1}		Observations		
		ws < 4.2	4.2 < ws < 7.6	ws > 7.6
Model data	ws < 4.2	6265	2544	306
	4.2 < ws < 7.6	3454	4734	1937
	ws > 7.6	527	2034	8416
Number of points 30217 ; PC=64.3%; EBD=2.8%; POD1=61.1%; POD2=50.8%; POD3=79.0%				

Table 33: night2 - Contingency table for wind speed (ms^{-1}) in standard configuration

FATE - Monthly Report

wind direction degree	Observations			
	North	East	South	West
Model data	North	13830	789	49
	East	1110	2164	469
	South	118	522	3268
	West	448	100	74

Number of points 23560 ; PC=82.9%; EBD=1.1%; POD1=89.2%; POD2=60.5%; POD3=84.7%; POD4=42.0%

Table 34: night2 - Contingency table for wind direction (degree) in standard configuration

relative humidity percent	Observations		
	rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	5930	2028
	7.1 < rh < 15.3	4216	6234
	rh > 15.3	598	2832

Number of points 29917 ; PC=63.2%; EBD=2.1%; POD1=55.2%; POD2=56.2%; POD3=83.4%

Table 35: night2 - Contingency table for relative humidity (percent) in standard configuration

precipitable water vapor mm	Observations		
	pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	8021	1935
	1.9 < pwv < 3.5	1508	3853
	pwv > 3.5	84	954

Number of points 23508 ; PC=75.9%; EBD=0.5%; POD1=83.4%; POD2=57.1%; POD3=83.4%

Table 36: night2 - Contingency table for precipitable water vapor (mm) in standard configuration

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	3964	3239	1193
	0.6 < see < 0.8	2500	3273	2745
	see > 0.8	517	1135	3547
Number of points		22113	; PC=%; EBD=%; POD1=%; POD2=%; POD3=%	

Table 37: night2 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.0

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	6589	0	865
	0.6 < see < 0.8	0	7397	865
	see > 0.8	392	250	6620
Number of points		22113	; PC=%; EBD=%; POD1=%; POD2=%; POD3=%	

Table 38: night2 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.24

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	4063	2161	222
	$3.4 < \tau < 5.8$	1319	3641	2185
	$\tau > 5.8$	227	1931	4962
Number of points 20711 ; PC=61.2%; EBD=2.2%; POD1=72.4%; POD2=47.1%; POD3=67.3%				

Table 39: night2 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 0.0

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	5022	324	952
	$3.4 < \tau < 5.8$	360	6365	952
	$\tau > 5.8$	227	1044	6417
Number of points 20711 ; PC=86.0%; EBD=5.7%; POD1=89.5%; POD2=82.3%; POD3=77.1%				

Table 40: night2 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 1.22

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	3356	3420
	0.6 < glf < 0.7	2299	3096
	glf > 0.7	531	915
Number of points 21070 ; PC=38.6%; EBD=14.3%; POD1=54.3%; POD2=41.7%; POD3=22.5%			

Table 41: night2 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.0

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	5565	966
	0.6 < glf < 0.7	90	6423
	glf > 0.7	531	42
Number of points 21070 ; PC=79.4%; EBD=15.4%; POD1=90.0%; POD2=86.4%; POD3=48.9%			

Table 42: night2 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.14

4 DAY2: Contingency tables

Tables in section 4 are included in the range: 43-46.

wind speed ms^{-1}		Observations		
		ws < 4.6	4.6 < ws < 7.4	ws > 7.4
Model data	ws < 4.6	5406	1609	391
	4.6 < ws < 7.4	4435	5695	1865
	ws > 7.4	369	2491	8157
Number of points 30418 ; PC=63.3%; EBD=2.5%; POD1=52.9%; POD2=58.1%; POD3=78.3%				

Table 43: day2 - Contingency table for wind speed (ms^{-1}) in standard configuration

FATE - Monthly Report

wind direction degree	Observations			
	North	East	South	West
Model data	North	11127	955	34
	East	241	730	107
	South	69	98	1769
	West	3493	44	372

Number of points 26279 ; PC=76.2%; EBD=0.6%; POD1=74.5%; POD2=40.0%; POD3=77.5%; POD4=88.3%

Table 44: day2 - Contingency table for wind direction (degree) in standard configuration

relative humidity percent	Observations		
	rh < 8.9	8.9 < rh < 17.9	rh > 17.9
Model data	rh < 8.9	7183	1761
	8.9 < rh < 17.9	3714	6798
	rh > 17.9	285	2434

Number of points 30343 ; PC=66.8%; EBD=1.1%; POD1=64.2%; POD2=61.8%; POD3=76.8%

Table 45: day2 - Contingency table for relative humidity (percent) in standard configuration

precipitable water vapor mm	Observations		
	pwv < 2.1	2.1 < pwv < 3.7	pwv > 3.7
Model data	pwv < 2.1	8075	1402
	2.1 < pwv < 3.7	1840	3940
	pwv > 3.7	100	1062

Number of points 23697 ; PC=76.4%; EBD=0.5%; POD1=80.6%; POD2=61.5%; POD3=83.7%

Table 46: day2 - Contingency table for precipitable water vapor (mm) in standard configuration

5 NIGHT3: Contingency tables

Tables in section 5 are included in the range: 47-56.

wind speed ms^{-1}		Observations		
		ws < 4.2	4.2 < ws < 7.6	ws > 7.6
Model data	ws < 4.2	5660	2738	415
	4.2 < ws < 7.6	4017	4472	2197
	ws > 7.6	607	2282	7829
Number of points 30217 ; PC=59.4%; EBD=3.4%; POD1=55.0%; POD2=47.1%; POD3=75.0%				

Table 47: night3 - Contingency table for wind speed (ms^{-1}) in standard configuration

FATE - Monthly Report

wind direction degree	Observations			
	North	East	South	West
Model data	North	13295	854	67
	East	1411	2113	589
	South	267	598	3103
	West	477	92	66

Number of points 23552 ; PC=79.4%; EBD=1.8%; POD1=86.1%; POD2=57.8%; POD3=81.1%; POD4=30.0%

Table 48: night3 - Contingency table for wind direction (degree) in standard configuration

relative humidity percent	Observations		
	rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	5564	2127
	7.1 < rh < 15.3	4504	5993
	rh > 15.3	626	2874

Number of points 29851 ; PC=60.4%; EBD=2.8%; POD1=52.0%; POD2=54.5%; POD3=79.4%

Table 49: night3 - Contingency table for relative humidity (percent) in standard configuration

precipitable water vapor mm	Observations		
	pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	8069	2128
	1.9 < pwv < 3.5	1249	3565
	pwv > 3.5	223	1056

Number of points 23323 ; PC=73.2%; EBD=1.4%; POD1=84.6%; POD2=52.8%; POD3=77.4%

Table 50: night3 - Contingency table for precipitable water vapor (mm) in standard configuration

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	3960	3225	1301
	0.6 < see < 0.8	2362	3060	2638
	see > 0.8	665	1377	3492
Number of points		22080	; PC=%; EBD=%; POD1=%; POD2=%; POD3=%	

Table 51: night3 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.0

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	6493	0	904
	0.6 < see < 0.8	0	7382	904
	see > 0.8	494	280	6527
Number of points		22080	; PC=%; EBD=%; POD1=%; POD2=%; POD3=%	

Table 52: night3 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.24

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	3835	2116	231
	$3.4 < \tau < 5.8$	1388	3474	1876
	$\tau > 5.8$	348	2203	5258
Number of points 20729 ; PC=60.6%; EBD=2.8%; POD1=68.8%; POD2=44.6%; POD3=71.4%				

Table 53: night3 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 0.0

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	4824	271	898
	$3.4 < \tau < 5.8$	399	6337	898
	$\tau > 5.8$	348	1185	6467
Number of points 20729 ; PC=85.0%; EBD=6.0%; POD1=86.6%; POD2=81.3%; POD3=78.3%				

Table 54: night3 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 1.22

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	2999	3020
	0.6 < glf < 0.7	2410	3451
	glf > 0.7	758	947
Number of points 21022 ; PC=39.5%; EBD=14.9%; POD1=48.6%; POD2=46.5%; POD3=25.0%			

Table 55: night3 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.0

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	5313	833
	0.6 < glf < 0.7	96	6553
	glf > 0.7	758	32
Number of points 21022 ; PC=79.5%; EBD=15.9%; POD1=86.2%; POD2=88.3%; POD3=51.2%			

Table 56: night3 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.14

6 DAY3: Contingency tables

Tables in section 6 are included in the range: 57-60.

wind speed ms^{-1}		Observations		
		ws < 4.6	4.6 < ws < 7.4	ws > 7.4
Model data	ws < 4.6	5146	1678	377
	4.6 < ws < 7.4	4578	5632	1974
	ws > 7.4	423	2475	8068
Number of points 30351 ; PC=62.1%; EBD=2.6%; POD1=50.7%; POD2=57.6%; POD3=77.4%				

Table 57: day3 - Contingency table for wind speed (ms^{-1}) in standard configuration

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wind direction degree	Observations			
	North	East	South	West
Model data	North	10741	978	51
	East	280	704	95
	South	126	78	1695
	West	3773	63	440

Number of points 26251 ; PC=73.6%; EBD=0.9%; POD1=72.0%; POD2=38.6%; POD3=74.3%; POD4=85.7%

Table 58: day3 - Contingency table for wind direction (degree) in standard configuration

relative humidity percent	Observations		
	rh < 8.9	8.9 < rh < 17.9	rh > 17.9
Model data	rh < 8.9	6807	1663
	8.9 < rh < 17.9	3961	6640
	rh > 17.9	417	2650

Number of points 30276 ; PC=64.4%; EBD=1.7%; POD1=60.9%; POD2=60.6%; POD3=74.5%

Table 59: day3 - Contingency table for relative humidity (percent) in standard configuration

precipitable water vapor mm	Observations		
	pwv < 2.1	2.1 < pwv < 3.7	pwv > 3.7
Model data	pwv < 2.1	7879	1691
	2.1 < pwv < 3.7	1996	3584
	pwv > 3.7	120	1082

Number of points 23618 ; PC=73.6%; EBD=0.8%; POD1=78.8%; POD2=56.4%; POD3=81.4%

Table 60: day3 - Contingency table for precipitable water vapor (mm) in standard configuration